

MuJoCo Left Panel Guide

This guide explains the left-side controls visible when the AMR simulation opens, and how to use the black **AMR CONTROL PANEL** overlay for day-to-day monitoring.

Use the left red/black MuJoCo menu for simulator control. Use the black AMR overlay for robot status, autonomy status, and live ROS topics.

1. What You Are Looking At

MuJoCo Left Menu

This is the built-in simulator control panel. It lets you save, pause, reset, inspect sensors, and tune visualization behavior.

AMR Control Panel

This is our custom overlay. It shows the robot battery, dock state, charging state, autonomy state, current command source, and key ROS topics.

2. MuJoCo Left Menu

File

- **Save xml** : exports the currently loaded model XML.
- **Save mjb** : saves the compiled MuJoCo binary model.
- **Print model** : prints model structure information to the console.
- **Print data** : prints current simulation state data.
- **Screenshot** : saves the current viewer image.
- **Quit** : closes the viewer.

Option

- **Help** : basic MuJoCo interaction help.
- **Info** : current viewer/model information.
- **Profiler** : performance timing view.
- **Sensor** : sensor inspection view.
- **Pause update** : stops screen updates while simulation may continue.
- **Fullscreen** : expands the viewer window.
- **Vertical Sync** : can reduce tearing.
- **Busy Wait** : timing/debug option; usually leave it alone.
- **Spacing**, **Color**, **Font** : visual appearance of the MuJoCo UI panel.

Simulation

- **Pause** : freezes simulation progress.
- **Run** : resumes simulation.

- **Reset** : restarts the scene from the initial state.
- **Reload** : reloads the model if the source changed.
- **Align** : re-aligns solver state.
- **Copy state** : copies the current sim state internally.
- **Key / Load key / Save key** : state snapshot utilities.
- **Noise scale** , **Noise rate** : simulator noise/debug fields.
- **History** : history buffer setting for simulation traces.

Watch, Physics, Rendering, Visualization, Group enable

- These sections are advanced MuJoCo tools for debugging internal state.
- For normal AMR operation, you usually do not need to expand them.
- Use them only if you want low-level physics or render diagnostics.

3. AMR Control Panel

Battery

Shows current simulated battery percentage.

- Near **100%** : robot is fully charged or recently started.
- Near **20%** : robot should switch to dock-seeking mode automatically.

Dock Contact

Shows whether the robot is physically aligned and touching the docking area.

- **NO** : robot is away from dock.
- **YES** : robot reached dock contact zone.

Charging

Shows whether charging is actively happening.

- **NO** : robot is not charging yet.
- **YES** : dock contact is good and charging logic is active.

Autonomy

Shows the current autonomy state.

- **PATROL** : robot is doing automatic movement through waypoints.
- **SEEK_DOCK** : battery is low, robot is returning to the dock.
- **FINAL_DOCKING** : robot is doing fine alignment at the dock.
- **CHARGING** : robot is docked and charging.
- **IDLE** : robot is waiting or no motion source is active.

Cmd Source

Shows which velocity source is currently driving the robot.

- **auto** : autonomy manager is driving the AMR.

- `manual` : keyboard teleop is overriding autonomy.
- `nav` : Nav2 is commanding the robot.
- `r1` : RL policy node is commanding the robot.
- `idle` : no active source.

Linear Cmd / Angular Cmd

Shows live command velocities currently applied to the AMR.

- `Linear Cmd` : forward/backward motion in meters per second.
- `Angular Cmd` : turning motion in radians per second.

4. Sensor List In The Overlay

- `/scan` : fused lidar scan used for general obstacle awareness.
- `/scan/front_left` : front-left lidar raw scan.
- `/scan/rear_right` : rear-right lidar raw scan.
- `/camera/depth/points` : depth camera point cloud.
- `/odometry/filtered` : EKF-fused odometry.
- `/battery_state` : ROS battery topic.

5. Normal Workflow

1. Launch the full stack and wait until the MuJoCo window opens.
2. Look at `Autonomy` and `Cmd Source`.
3. If autonomy is healthy, the robot should enter patrol motion automatically.
4. Watch battery percentage drop slowly during movement.
5. At low battery, the robot should change to `SEEK_DOCK` and go toward the charger.
6. When contact is achieved, `Dock Contact` should become `YES`.
7. Then `Charging` should become `YES`.

6. Teleop Override

Use this command in another terminal:

```
ros2 run teleop_twist_keyboard teleop_twist_keyboard --ros-args --remap cmd_vel:=/cmd_vel_manual
```

- If teleop is active, `Cmd Source` should change to `manual`.
- Manual commands have priority over autonomy.
- When you stop sending keys, control can return to autonomy after timeout.

7. Troubleshooting From The Panel

If the robot is visible but not moving, first check `Autonomy`, then `Cmd Source`, then the command values.

- `Autonomy = IDLE` : autonomy is not active or no valid pose/scan data is available.

- `Cmd Source = manual` with zero commands: a teleop source may still be overriding.
- `Battery = 100%` and no movement: check that the sim node did not crash in the launch terminal.
- `SEEK_DOCK` all the time: battery may be below threshold or docking logic latched.
- `Dock Contact = YES` but `Charging = NO`: robot may still be slightly rotating or misaligned.

8. Recommended Daily Use

- Keep the MuJoCo left menu visible.
- Do not press `Reset` while testing a docking sequence unless you want a full restart.
- Use `Screenshot` when documenting behavior.
- Use the AMR overlay as the first quick health check before opening RViz.

Document for: `mujoco_amr_sim`

Generated inside the project workspace for quick operator reference.